



# Vidyavaridhi's College of Engineering & Technology

Department of Computer Engineering

Academic Year: 2024-25

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|-----------------------------------|
| Experiment No. 7                  |
| Implement Time Series Forecasting |
| Date of Performance: 17/03/2025   |
| Date of Submission: 24/03/2025    |



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**Aim:** Implement Time Series Forecasting.

**Objective:** Able to perform Time Series Forecasting.

## Theory:

Time series forecasting is employed in various sectors, including finance, supply chain management, production, and inventory planning, making it one of the most widely used data science approaches. Time series forecasting has many applications, including resource allocation, business planning, weather forecasts, and stock price prediction.

## Time series forecasting modeling

There are many techniques used in time series forecasting that try to achieve precision and reduce mistakes and losses. Still, several classical and contemporary machine learning techniques have demonstrated their efficacy and computational use. We shall discuss several different forms of time series analysis below.

## Best machine learning models for time series forecasting

For time series forecasting, a variety of models can be utilized. For instance, the LSTM Network is a unique type of neural network that makes predictions based on historical data. It is widely used for many things, including time series analysis and language recognition. By adding a series of delays to the input, models like the random forest, gradient boosting regressor, and time delay neural networks may include temporal information and represent the data at various periods in time. TDNNs are built as feedforward neural networks rather than recurrent neural networks because of their sequential nature.

## Code:

[https://colab.research.google.com/drive/1eQQxhlZ8Pqludor92hf\\_OhAgUlbeRorI?usp=sharing](https://colab.research.google.com/drive/1eQQxhlZ8Pqludor92hf_OhAgUlbeRorI?usp=sharing)

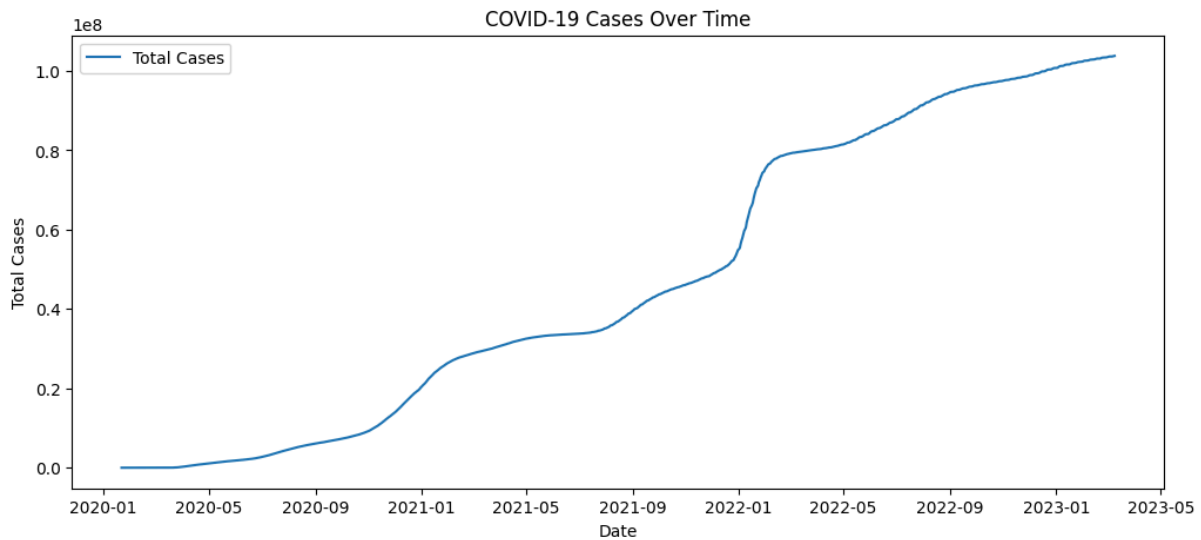


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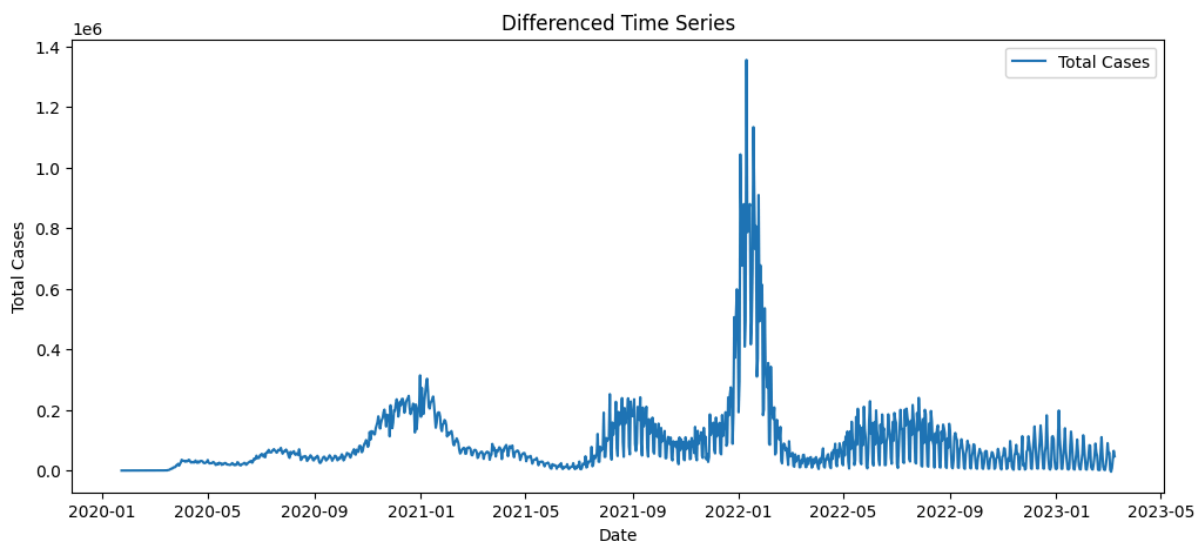
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## Output:



```
ADF Statistic: -0.03337976177101878  
p-value: 0.9556829980610124  
Series is NOT stationary
```



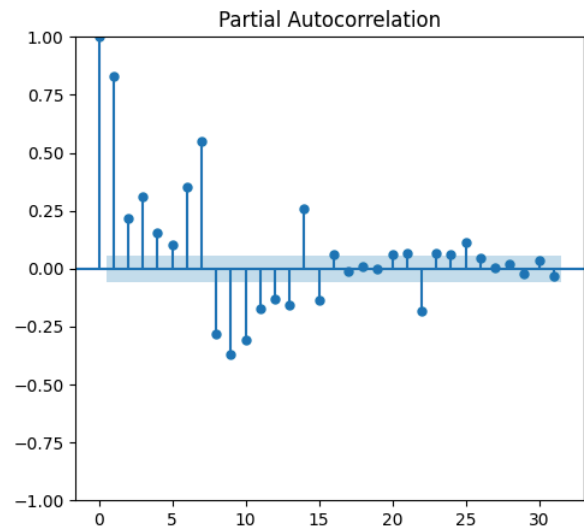
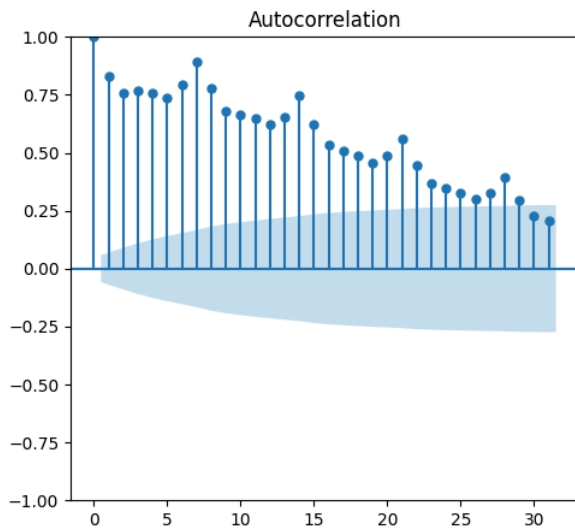


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ADF Statistic: -4.326191879161942  
p-value: 0.0004000393176799628  
Series is stationary



```
SARIMAX Results
=====
Dep. Variable:      y      No. Observations:      1143
Model:              ARIMA(2, 1, 2)      Log Likelihood      -14229.822
Date:               Thu, 03 Apr 2025      AIC      28469.645
Time:               15:52:50      BIC      28494.847
Sample:             01-22-2020      HQIC      28479.161
                  - 03-09-2023
Covariance Type:    opg
=====
              coef      std err      z      P>|z|      [0.025      0.975]
-----
ar.L1          0.4788        0.033     14.525      0.000        0.414        0.543
ar.L2          0.5090        0.033     15.429      0.000        0.444        0.574
ma.L1         -0.0196        0.024     -0.816      0.415       -0.067        0.027
ma.L2         -0.5332        0.017    -32.253      0.000       -0.566       -0.501
sigma2        3.966e+09      2.89e-12     1.37e+21     0.000      3.97e+09      3.97e+09
=====
Ljung-Box (L1) (Q):      0.00      Jarque-Bera (JB):      80181.38
Prob(Q):                 0.97      Prob(JB):              0.00
Heteroskedasticity (H):   7.74      Skew:                  1.95
Prob(H) (two-sided):      0.00      Kurtosis:              43.86
=====
```

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

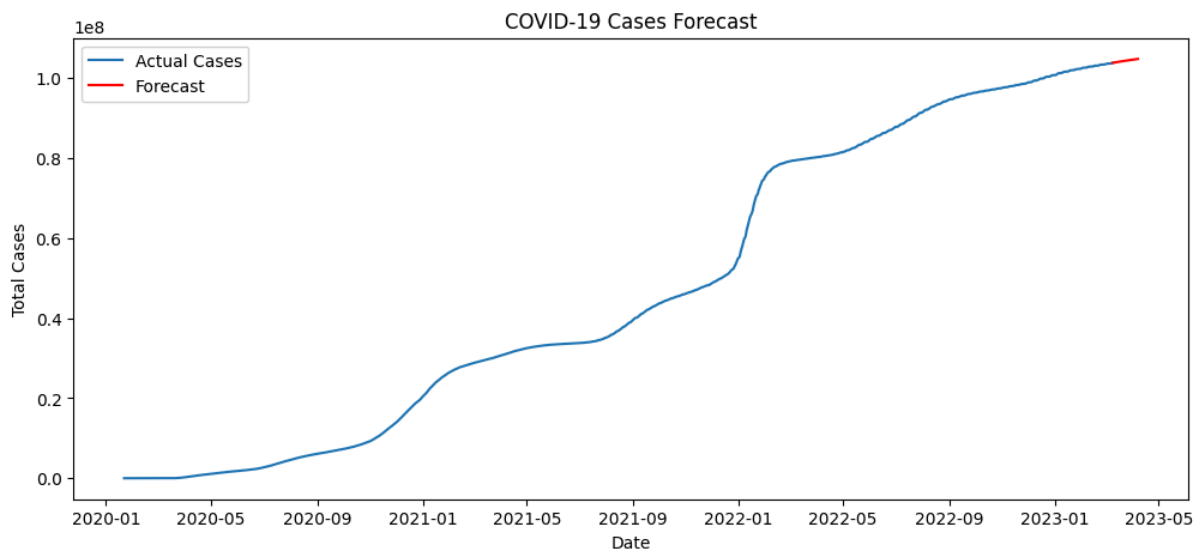
[2] Covariance matrix is singular or near-singular, with condition number 5.12e+36. Standard errors may be unstable.



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**Conclusion:** The results obtained from time series forecasting provide valuable insights into future trends based on historical data patterns. By analyzing the predicted values, we can assess the accuracy and reliability of the model in capturing underlying seasonality, trends, and fluctuations. If the forecast closely aligns with actual observations, it indicates that the model is well-tuned and effective. However, deviations between predicted and actual values may suggest the need for further model refinement, such as adjusting parameters, incorporating additional variables, or using more advanced forecasting techniques. Overall the forecasting results help in making informed decisions, optimizing resource planning, and identifying potential risks or opportunities in the given dataset.